

4.7 Acid/base equilibria

During this topic there will be the opportunity to carry out an internal assessment activity. Please see *Appendix 9* for more details.

Students will be assessed on their ability to:

- a demonstrate an understanding that the theory about acidity developed in the 19th and 20th centuries from a substance with a sour taste to a substance which produces an excess of hydrogen ions in solution (Arrhenius theory) to the Brønsted-Lowry theory
- b demonstrate an understanding that a Brønsted-Lowry acid is a proton donor and a base a proton acceptor and that acid-base equilibria involve transfer of protons
- c demonstrate understanding of the Brønsted-Lowry theory of acid-base behaviour, and use it to identify conjugate acid-base pairs
- d define the terms pH, K_a and K_w , pK_a and pK_w , and be able to carry out calculations relating the pH of strong acids and bases to their concentrations in mol dm^{-3}
- e demonstrate an understanding that weak acids and bases are only slightly dissociated in aqueous solution, and apply the equilibrium law to deduce the expressions for the equilibrium constants K_a and K_w